SHAMRAT NEERO

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GitHub | LinkedIn | Portfolio

Summary Computer science graduate with research and engineering experience in trustworthy medical AI, neuroimaging, and multimodal learning. Research spans chest X-ray triage, MRI–genetics fusion, external validation, model calibration, and deployment-aware evaluation.

Education

Bachelor of Science in Computer Science, BRAC University, Bangladesh CGPA: 3.06/4.00

Undergraduate thesis: CyberGuard Chronicles: Design, Implementation, and Assessment of a Game-Based Platform for Cybersecurity Awareness in Bangladesh (Thesis GPA: 4.00).

Publications & Manuscripts

• Neero, S. "AI-Assisted Chest X-Ray Triage: An Integrated Evaluation of Ranking, Localization, and Calibration for Deployment in Resource-Constrained Settings." MICAD 2026, accepted.

• Neero, S. "Checkpoint selection by validation AUC can silently produce degenerate models in multimodal Alzheimer's disease classification—a cross-cohort audit." Submitted to Computerized Medical Imaging and Graphics (CMIG), 2026.

Technical Skills

Machine Learning: Python, PyTorch, scikit-learn, TorchVision, OpenCV, 3D CNNs, Grad-CAM, calibration, subject-level cross-validation.

Imaging & Data: Structural MRI, DICOM, FastSurfer, dcm2niix, ADNI, OASIS-3, NIH ChestX-ray14, VinDr-CXR, SQL.

Engineering: FastAPI, Streamlit, Django, React, JavaScript, Docker, WSL2, Git, PostgreSQL.

Research Projects

AI-Assisted Chest X-Ray Triage for Resource-Constrained Settings — MICAD 2026 (Accepted) 2026

PyTorch, DenseNet-121, ConvNeXt-Tiny, Grad-CAM, FastAPI, Streamlit

• Solo-authored study evaluating four deep learning models on 112K+ NIH ChestX-ray14 images, with external validation on VinDr-CXR.

• Demonstrated disagreement among ranking (AUC), localization, and calibration; reported Cardiomegaly localization IoU up to 0.47 and approximately 40-fold lower ECE using per-class isotonic calibration versus global temperature scaling.

• Built and benchmarked a FastAPI + Streamlit triage prototype, profiling CPU/GPU latency and memory for resource-constrained deployment.

Alzheimer's Disease Classification: MRI + APOE Genetics Fusion 2026

PyTorch, 3D CNN, OASIS-3, ADNI, external validation

• Evaluated clinical, imaging, and fusion models on OASIS-3 (365 subjects, five-fold subject-level cross-validation), followed by external evaluation on ADNI (1,287 participants).

• Audited discrepancies between mean fold-wise and pooled out-of-fold AUC; identified degenerate AUC-selected checkpoints and cross-fold probability incoherence.

• Using validation Brier score for checkpoint selection improved external fusion AUC from 0.6330 to 0.8014 in the cross-cohort evaluation.

ADNI MCI 24-Month Progression — Longitudinal Cohort Study 2026 (ongoing)

Structural T1 MRI, ADNI, FastSurfer, longitudinal outcome modeling

• Defined a baseline MCI cohort of 1,744 participants: 196 converters, 829 stable MCI, and 719 right-censored, with explicit 24-month outcome rules.

• Selected 954 baseline T1 MRI studies for imaging processing; developing MRI-derived and clinical predictor pipelines. Imaging analysis and progression modeling are ongoing.

Bangla Sign Language Recognition — Research + Full-Stack Deployment 2025

ResNet18, PyTorch, OpenCV, Django, React

• Achieved 97% reported accuracy on BDSL-49 with ResNet18 transfer learning; developed a full-stack webcam inference application. Manuscript submitted to ICCIT.

Other Research & Engineering

CyberGuard Chronicles — Undergraduate Thesis 2025

• Built a game-based cybersecurity-awareness intervention and evaluated it through semi-structured interviews with rural and urban participants; thesis GPA: 4.00.

Web Developer — THiNK LABS Internship

• Developed React interfaces and contributed to COMSURE power-management software; built a JavaScript pixel game.